

A PROJECT Title

ON

" A Web-Based Patient Registration and Electronic Medical Record Management Portal "

Insem-End project submitted in partial fulfilment of the requirements for the Course

Front End Development Frameworks *for the degree of*

BACHELOR OF TECHNOLOGY

IN

Computer Science and Engineering

(2025-2026 A.Y)

BY

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title: A Web-Based Patient Registration and Electronic Medical Record Management Portal

1. Abstract:

The Patient Registration and Electronic Medical Record (EMR) Portal is a web-based healthcare management system developed to simplify and digitize patient onboarding and medical record management. Traditional hospital registration processes often involve paper-based documentation, manual verification, and difficulty in maintaining accurate patient records. This project provides an efficient digital solution for managing patient information securely and systematically.

The portal enables patients to register online by entering personal details, medical history, allergies, current medications, insurance details, and emergency contact information. It also allows uploading profile photos and completing digital consent forms. The system ensures organized storage of all patient-related data, making it easier for healthcare administrators to access and manage records.

The application is developed using React.js for the frontend to create an interactive and responsive user interface, while Node.js and Express.js are used for backend API development. REST APIs facilitate communication between frontend and backend components. Data can be stored using MySQL or Local Storage for basic implementation. Axios is used for API requests, and Postman is utilized for testing API endpoints.

The system includes multiple modules such as patient registration, medical history management, allergy tracking, medication records, emergency contacts, insurance information, consent form submission, admin dashboard, profile management, and API integration. React Router enables seamless navigation, while Context API or Redux manages application state efficiently.

This project improves healthcare workflow by reducing paperwork, minimizing registration errors, improving accessibility of patient records, and enhancing operational efficiency. It demonstrates practical implementation of modern frontend development frameworks in the healthcare domain and provides a scalable foundation for advanced hospital management systems.

2. INTRODUCTION:

Healthcare institutions require accurate and efficient patient data management to provide quality medical services. Traditional manual registration systems are time-consuming, prone to human error, and difficult to maintain.

The Patient Registration & EMR Portal is designed to digitize patient onboarding and medical record handling. It enables patients to submit details online, reducing waiting time and improving hospital efficiency.

This project focuses on building a secure and responsive healthcare portal using modern web technologies. It allows management of patient profiles, medical history, allergy information, medication records, emergency contacts, insurance details, and digital consent forms.

The system demonstrates the practical application of frontend development frameworks and RESTful API integration in healthcare solutions.

3. SOFTWARE REQUIREMENTS:

Frontend

- HTML5
- CSS3
- JavaScript (ES6+)
- React.js
- React Router
- Axios
- Bootstrap / Tailwind CSS

Backend

- Node.js
- Express.js

Database

- MySQL (Optional)
- Local Storage

API Testing

- Postman

State Management

- Context API / Redux

Development Tools

- Visual Studio Code
- npm

Browser Support

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

Modules Implemented

1. Patient Registration Form
2. Medical History Module
3. Allergy Information Input
4. Current Medication List
5. Emergency Contact Management
6. Consent Form Submission
7. Profile Photo Upload
8. Insurance Details Management
9. Admin Dashboard
10. Postman API Testing Module

Git Hub:

Sahithya: <https://github.com/sahithyagundeboina>

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